

Technical Datasheet TnH Sensor Modules with Modbus RTU Protocol with RS485 Interface

The SN modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (this feature however also depends on the signal strength of the Modbus Master Device).

The RS485 SN-TnH module is sturdy, has low power usage and is easy to use.

TnH Sensor Module: -



The SN modules can be mounted in Outdoor or Indoor environment. **Soft setting** for **Slave ID** is available in the software. In Addition, it has reset switch for Slave ID reset if forgotten.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power and Communication** port.

Specifications

General –

Dimensions	158mm L x 139 mm D
Power	Input Power – 12 - 24 V DC
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



Configuration Settings: -

Communication Speed	9600
Data Bits	8
Parity	None
CRC	Yes
Slave ID	Software setting(1-247)

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	Temperature	40000	0X03	RS485	16 Bit Unsigned int
2	Humidity	40001	0X03	RS485	16 Bit Unsigned int
3	Current Slave ID	40002	0X03	RS485	16 Bit Unsigned int
4	Enter New Slave ID	40003	0X03	RS485	16 Bit Unsigned int

Note: -

For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

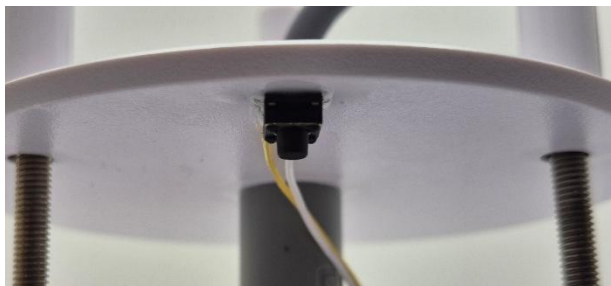
22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

- Here is an example of receiving data on Mb poll software.
- Open Mb-poll software connect using configuration setting (Fixed baud rate is 9600 and Default slave id is 1).
- After connecting you will get data as shown below

Tx = 129: Err = 104: ID = 1: F = 03: SR = 1000ms

	Alias	00000
0	Temperature	3018
1	Humidity	4244
2	Current Slave ID	1
3	New Slave ID	0
4		

- As you can see in the image above, the resistor 40002 –1 represents the present slave ID of the module
- To change the slave id, you must enter a value on the 40003 resistor for the new slave id as shown in the image.
- After that the slave ID will be changed.
- Somehow you forgot the slave id of the board, in that case press reset button and remove power supply. Then you will get the default slave ID 1 for the system.



Write Single Register

Slave ID: 1

Address: 3

Value: 6

Send

Cancel

Result

N/A

☒ Close dialog on "Response ok"

Use Function

☒ 06: Write single register
 ☐ 16: Write multiple registers

Sensor Name	Temperature Measuring Range	Humidity Measuring range	Operating Temperature	Measuring Accuracy
AHT20	-20°C ~ 85°C.	0~100% RH.	-40°C ~ 120°C.	0.3°C, 2.0 %RH
AHT21	-40°C ~ 120°C.	0~100% RH.	-40°C ~ 120°C.	0.3°C, 2.0 %RH

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